

Sentiment Analysis

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The Little Prince example

“This is a nice book for both young and old. It gives beautiful life lessons in a fun way. Definitely worth the money!”

+ Educational

+ Fun

+ Price

“Nice story for older children.”

+ Funny

- Readability

ANTOINE DE SAINT-EXUPÉRY

The Little Prince

The original
translation by
Katherine Woods,
with full-colour
illustrations



Conceptual challenges

Which of the following sentences express sentiment? If so, which is their polarity (positive, neutral or negative)?

- There was an earthquake in Japan
- They said it would be a great experience...
 - ...And yes it was!
 - ...However, I didn't have fun.
- Go read the book
- Yeah, sure!

Sentiment

- **Sentiment** =
 - Feelings, Attitudes, Emotions
 - A thought, view, or attitude, especially one based mainly on emotion instead of reason
 - Subjective impressions, not facts
- **Opinion** = a sentiment expressed about a specific target (more on this later)

Sentiment analysis

- Use of natural language processing (NLP) and computational techniques to automate the extraction or classification of sentiment from unstructured text
- Other terms (often used interchangeably)
 - Sentiment classification
 - Sentiment mining
 - **Opinion mining**: emphasizes the fuller structure of *who* holds *what* sentiment, about *what* target, and *when* (see “Practical definition” later)

Related tasks

- Emotion detection (e.g., happiness, anger, sadness)
- Stance detection (in favor, neutral or against)
- Subjectivity analysis (objective vs subjective text)
- Sarcasm/Irony detection
- Hate-speech detection

Sentiment analysis

- Can be applied to almost every topic & domain (non exhaustive list):
 - Book: is this review *positive* or *negative*?
 - Products: what do people *think about* the new iPhone?
 - Blog: how are people *thinking about* immigrants?
 - Politics: *who* is going to win the election?
 - Social Media: what is the *trend* today?
 - Movie: is this review *positive* or *negative* (IMDB, Netflix)?
 - Marketing: how is consumer *confidence*? Consumer *attitudes*?
 - Healthcare: are patients *happy* with the hospital environment?

Opinion types

- **Regular opinions:** an evaluation of a target entity
 - e.g., “The *touch screen* is really cool.”
- **Comparative opinions:** Comparison of more than one entity.
 - e.g., “iPhone *is better than* Blackberry.”

Opinion types

- **Regular opinions:** an evaluation of a target entity, stated either explicitly (direct) or implied through a statement (indirect)
 - **Direct opinions:**
 - * e.g., “The *touch screen* is really cool.”
 - **Indirect opinions:**
 - * e.g., “After taking *the drug*, my pain has gone.”
 - Positive or negative? About what/whom?
- **Comparative opinions:** Comparison of more than one entity.
 - e.g., “iPhone *is better than* Blackberry.”

Practical definition

- An **opinion** is a quintuple:
 - *entity*: target entity (or object).
 - *aspect*: aspect (or feature) of the entity.
 - *sentiment*: +, -, or neutral, a rating, or an emotion.
 - *holder*: opinion holder.
 - *time*: time when the opinion was expressed.

Revisiting The Little Prince example

“This is a nice book for both young and old. It gives beautiful life lessons in a fun way. Definitely worth the money!”

+ Educational

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Sentiment analysis tasks

- Simplest task:

- Is this text positive or negative?
- **More complex:**
 - Is this text positive, negative or neutral?
 - Label this text from 1 to 5
- **Advanced:**
 - Detect the entity, aspect, sentiment, holder and time
 - Implicit opinions

Document sentiment analysis

- Classify a document (e.g., a review) based on the overall sentiment of the opinion holder
 - Classes: Positive, negative, and neutral
 - Essentially a text classification problem.
- An example review:
 - *“I bought an iPhone a few days ago. It is such a nice phone, although a little large. The touch screen is cool. The voice quality is great too. I simply love it!”*

Sentence sentiment analysis

- Classify the sentiment expressed in a sentence
 - Classes: positive, negative, and neutral
- But bear in mind:
 - *Explicit opinion*: “I like this car.”
 - *Implied (indirect) opinion*: “I bought this car yesterday and it broke today.”
 - *Mixed opinion*: “Apple is doing well in this poor economy”

Challenges

- What challenges do you think we will face in sentiment analysis?

Challenges

- Hard to do with *bag of words*
- Must consider other features due to...
 - Subtlety of sentiment expression
 - * irony (*“What a great car, it stopped working on the second day.”*)
 - * expression of sentiment using neutral words (*“The concert didn’t meet my expectations.”*)
 - Domain/context dependence
 - * words/phrases can mean different things in different contexts and domains (*“long queue” vs “long battery life”*)
 - Effect of syntax on semantics (*Negation*)

Methods for sentiment analysis

- **Lexicon-based methods**
 - Dictionary based: Using sentiment words and phrases (e.g., “good”, “wonderful”, “awesome”, “troublesome”, “cost an arm and leg”)
 - Corpus-based: Using co-occurrence statistics or syntactic patterns embedded in text corpora
- **Supervised learning methods:** to classify reviews into positive and negative.
 - Traditional Machine Learning: Naïve Bayes, Support Vector Machine
 - Deep learning: pretrained language models fine-tuned on labeled sentiment data (e.g., BERT, GPT)
- **Prompt-based methods:**

- Use large language models (LLMs) to classify sentiment by providing a prompt and asking the model to generate a response.
- More to come later in the course.

Lexicon-based Methods

Sentiment and other lexicons

- Lists of words that are associated with sentiment scores
- Can have binary scores (1, -1) or intensity scores (from 0 to 1)
- Positive/negative polarity, emotions, affective states, negation lists

brainwashing	-3
brave	2
breakthrough	3
breathtaking	5
bribe	-3
bright	1
brightest	2
brightness	1
brilliant	4
brisk	2
broke	-1
broken	-1

Basic Lexicon Approach

- Detect sentiment in two independent dimensions:
 - Positive: {1, 2, ... 5}
 - Negative: {-5, -4, ... -1}
- Example: “He is brilliant but boring”
 - Overall sentiment = ?

Basic Lexicon Approach

- Detect sentiment in two independent dimensions:
 - Positive: {1, 2, ... 5}
 - Negative: {-5, -4, ... -1}
- Example: “He is brilliant but boring”
 - Sentiment(“brilliant”) = +4
 - Sentiment(“boring”) = -2
 - Overall sentiment = +2
- Limitation: naive summation treats both words as equally important, but “but” signals that what comes after should dominate

LIWC (Linguistic Inquiry and Word Count) | Tausczik and Pennebaker (2010)

- 2,300 words, >70 classes, based on psychological theory and research
- **Affective Processes**

- negative emotion (“bad”, “weird”, “hate”, “problem”, “tough”)
- positive emotion (“love”, “nice”, “sweet”)
- **Cognitive Processes**
 - Tentative (“maybe”, “perhaps”, “guess”), Inhibition (“block”, “constraint”)
- **Pronouns, Negation** (“no”, “never”), **Quantifiers** (“few”, “many”)

VADER Sentiment Analysis | Hutto and Gilbert (2014)

- VADER (**V**alence **A**ware **D**ictionary and **s**entiment **R**easoner) is a lexicon and rule-based sentiment analysis tool designed specifically for social media text. Contains a pre-built lexicon of words that are associated with sentiment scores ranging from -4 to +4
- Five generalizable heuristics based on grammatical and syntactical cues:
 - **Punctuation**: “The food here is good **!!!**” vs “The food here is good.”
 - **Capitalization**: “The food here is **GREAT!**” vs “The food here is great!”
 - **Degree modifiers**: “The service here is **extremely** good” vs “The service here is good”
 - **The conjunction “but”**: “The food here is great, **but** the service is horrible” has mixed sentiment
 - **For negation**: examine the tri-gram preceding a sentiment lexical feature: “The food **here isn’t that** great”

SentiWordNet | Esuli and Sebastiani (2006)

- <https://github.com/aesuli/SentiWordNet>
- All WordNet synsets automatically annotated for degrees of positivity, negativity, and objectivity
- **estimable(J,1)**: “deserving of respect or high regard”

Pos .75 Neg 0 Obj .25

- **estimable(J,3)**: “may be computed or estimated”

Pos 0 Neg 0 Obj 1

How to measure polarity of a phrase?

- Positive phrases co-occur more with “excellent”
- Negative phrases co-occur more with “poor”
- But how to measure co-occurrence?

Pointwise Mutual Information

- **PMI between two words**:
 - How much more do two words co-occur than if they were independent?
- **Positive PMI (> 0)**: x and y co-occur more often than expected.
- **PMI = 0**: They are independent.
- **Negative PMI (< 0)**: They co-occur less often than expected.

$$PMI(word_1, word_2) = \log_2 \frac{P(word_1, word_2)}{P(word_1)P(word_2)}$$

How to estimate PMI

$$PMI(word_1, word_2) = \log_2 \frac{P(word_1, word_2)}{P(word_1)P(word_2)}$$

- $P(\text{word})$ estimated by $\text{hits}(\text{word})/N$
- $P(\text{word}_1, \text{word}_2)$ by $\text{hits}(\text{word}_1 \text{ NEAR } \text{word}_2)/N^2$

How to estimate PMI

$$PMI(word_1, word_2) = \log_2 \frac{\text{hits}(word_1 \text{ NEAR } word_2)}{\text{hits}(word_1)\text{hits}(word_2)}$$

Does phrase appear more with “poor” or “excellent”?

$$\text{Polarity}(\textit{phrase}) = \text{PMI}(\textit{phrase}, \text{“excellent”}) - \text{PMI}(\textit{phrase}, \text{“poor”})$$

Lexicon-based methods in summary

- **Intuition**
 - Start with a seed set of words (“good”, “poor”)
 - Find other words that have similar polarity:
 - * Using “and” and “but”
 - * Using words that occur nearby in the same document
 - * Using synonyms and antonyms
 - * Using rules based on punctuation, emoticons

Lexicon-based methods in summary (contd)

- What are the advantages and disadvantages?

Lexicon-based methods in summary (contd)

- **Advantages:**
 - Can be domain-independent with general purpose lexicons
 - Can become domain-dependent
 - Can be easy to interpret/explain prediction output
 - Can be applied when no training data is available
- **Disadvantages:**
 - Compared to a well-trained, in-domain ML model they typically underperform
 - Sensitive to dictionary coverage

Supervised Methods

Basic steps

- Pre-processing and tokenization
- Feature representation
- Feature selection
- Classification
- Evaluation

Sentiment tokenization issues

- Deal with HTML and XML markup
- X/Twitter mark-up (names, hash tags)
- Capitalization (preserve for words in all caps)
- Phone numbers, dates
- Emoticons
- Useful code:
 - Christopher Potts sentiment tokenizer
 - Brendan O’Connor twitter tokenizer

Potts emoticons

```

[<>]?           # optional hat/brow
[:;=8]          # eyes
[\-o\*\ ' ]?    # optional nose
[\)\]\]\(\[dDpP/\:}\{\@\\|\]\]  # mouth
|               ##### reverse orientation
[\)\]\]\(\[dDpP/\:}\{\@\\|\]\]  # mouth
[\-o\*\ ' ]?    # optional nose
[:;=8]          # eyes
[<>]?           # optional hat/brow

```

The danger of stemming

- The Porter stemmer identifies word suffixes and strips them off.
- But:
 - “objective” (pos) and “objection” (neg) -> “object”
 - “competence” (pos) and “compete” (neg) -> “compet”

Features for supervised learning

- The problem has been studied by numerous researchers.
- **Key:** feature engineering. A large set of features have been tried by researchers. E.g.,
 - Terms frequency and different weighting schemes
 - Part of speech (POS) tags
 - Opinion words and phrases
 - Negations
 - Stylistic
 - Syntactic dependency

Negation

Add NOT_ to every word between negation and following punctuation:

didn't like this movie , but I



didn't NOT_like NOT_this NOT_movie but I

Challenges of negation

- “terrible” vs “wasn't terrible”
 - *The movie was terrible*
 - *The movie was bad but wasn't that terrible as they said*
- The degree of the intensity shift varies from term to term for both positive and negative terms

Supervised sentiment analysis | Kiritchenko et al. (2014)

- A supervised statistical text classification approach based on surface, semantic, and sentiment features.
- For negation: estimate sentiment scores of individual terms in the presence of negation
- One lexicon for words in negated contexts and one for words in affirmative

Supervised sentiment analysis | Kiritchenko et al. (2014)

Feature group	Examples
word ngrams	<i>grrreat, show, grrreat.show, miss.NEG, miss.NEG.the</i>
character ngrams	<i>grr, grrr, grrre, rrr, rrre, rrrea</i>
all-caps	all-caps:1
POS	POS.N:1 (nouns), POS.V:2 (verbs), POS.E:1 (emoticons), POS,:1 (punctuation)
automatic lexicon features	HS.unigrams_positive_count:4, HS.unigrams_negative_total_score:1.51, HS.unigrams_POS.N_combined_total_score:0.19, HS.bigrams_positive_total_score:3.55, HS.bigrams_negative_max_score:1.98
manual lexicon features	MPQA_positive_affirmative_score:2, MPQA_negative_negated_score:1, BINGLIU_POS.V_negative_negated_score:1
punctuation	punctuation.!:1
emoticons	emoticon_positive:1, <i>emoticon_positive.last</i>
elongated words	elongation:1
clusters	<i>cluster_11111001110, cluster_10001111</i>

Table 6: Examples of features that the system would generate for message “GRRREAT show!!! Hope not to miss the next one :)”. Numeric features are presented in the format: <feature_name>:<feature_value>. Binary features are italicized; only features with value of 1 are shown.

Supervised sentiment analysis

- Advantages
 - Lead to better performance compared to lexicon based approaches
 - The output can be explained (most of the times)
- Disadvantages
 - They need training data
 - They can’t capture the context
 - Based on feature engineering that is a tedious task
 - Not good performance in multiclass classification

Deep Learning

Sentiment-specific word embedding | Tang et al. (2014)

- Word embedding models learn the syntactic context of words but ignore the sentiment of text
- Example:
 - “good” and “bad” will be represented as neighboring word vectors
 - “good” and “excellent” will be represented as more distant word vectors
- Solution: Learn sentiment specific word embedding

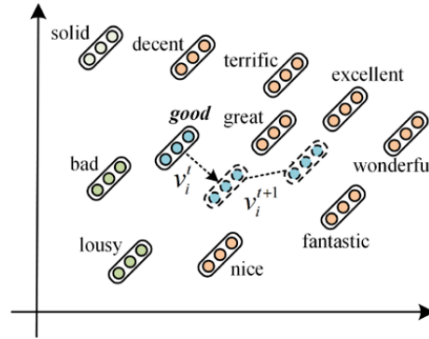
Word vector refinement | Yu et al. (2017)

- Start with a set of pre-trained word vectors and a sentiment lexicon

- Calculate the semantic similarity between each sentiment word and the other words in the lexicon based on the cosine distance of their pre-trained vectors
- Select top-k most similar words as the nearest neighbors and re-rank according to sentiment scores

Word vector refinement | Yu et al. (2017)

- Refine the pre-trained vector of the target word to be:
 - closer to its sentimentally similar neighbors,
 - further away from its dissimilar neighbors, and
 - not too far away from the original vector.



Sentiment analysis with BERT | Devlin et al. 2019

- Sentiment analysis was one of the tasks in the BERT paper

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4
OpenAI GPT	82.1/81.4	70.3	87.4	91.3
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9

Pre-trained models on SA

<https://huggingface.co/blog/sentiment-analysis-python>

- Twitter-roberta-base-sentiment is a roBERTa model trained on ~58M tweets and fine-tuned for sentiment analysis (<https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment>)
- SST-2 DistilBERT: Fine-tuned on the Stanford Sentiment Treebank (SST-2) which consists of sentences from movie reviews. The model is well-suited for general sentiment analysis tasks. (<https://huggingface.co/distilbert-base-uncased-finetuned-sst-2-english>)
- Bert-base-multilingual-uncased-sentiment is a model fine-tuned for sentiment analysis on product reviews in six languages: English, Dutch, German, French, Spanish and Italian (<https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment>)

- Distilbert-base-uncased-emotion is a model fine-tuned for detecting emotions in texts, including sadness, joy, love, anger, fear and surprise (<https://huggingface.co/bhadresh-savani/distilbert-base-uncased-emotion>)

Bias in Sentiment Analysis

Bias in sentiment analysis | Kiritchenko and Mohammad (2018)

- Examining Gender and Race Bias in Two Hundred Sentiment Analysis Systems (Kiritchenko & Mohammad, *SEM 2018)
- Are systems that detect sentiment biased?
- Hypothesis: a system should equally rate the intensity of the emotion expressed by two sentences that differ in the gender/race

Bias in sentiment analysis | Kiritchenko and Mohammad (2018)

Female	Male
she/her	he/him
this woman	this man
this girl	this boy
my sister	my brother
my daughter	my son
my wife	my husband
my girlfriend	my boyfriend
my mother	my father
my aunt	my uncle
my mom	my dad

African American		European American	
Female	Male	Female	Male
Ebony	Alonzo	Amanda	Adam
Jasmine	Alphonse	Betsy	Alan
Lakisha	Darnell	Courtney	Andrew
Latisha	Jamel	Ellen	Frank
Latoya	Jerome	Heather	Harry
Nichelle	Lamar	Katie	Jack
Shaniqua	Leroy	Kristin	Josh
Shereen	Malik	Melanie	Justin
Tanisha	Terrence	Nancy	Roger
Tia	Torrance	Stephanie	Ryan

Bias in sentiment analysis | Kiritchenko and Mohammad (2018)

- > 75% of systems mark one gender/race with higher intensity scores than other
- more widely prevalent for race than for gender
- impact on downstream applications?

Bias in sentiment analysis

What about biases in sentiment classification models?

- DistilBERT base uncased finetuned SST-2:
 - “This movie was filmed in France” -> ?
 - “This movie was filmed in Afghanistan” -> ?

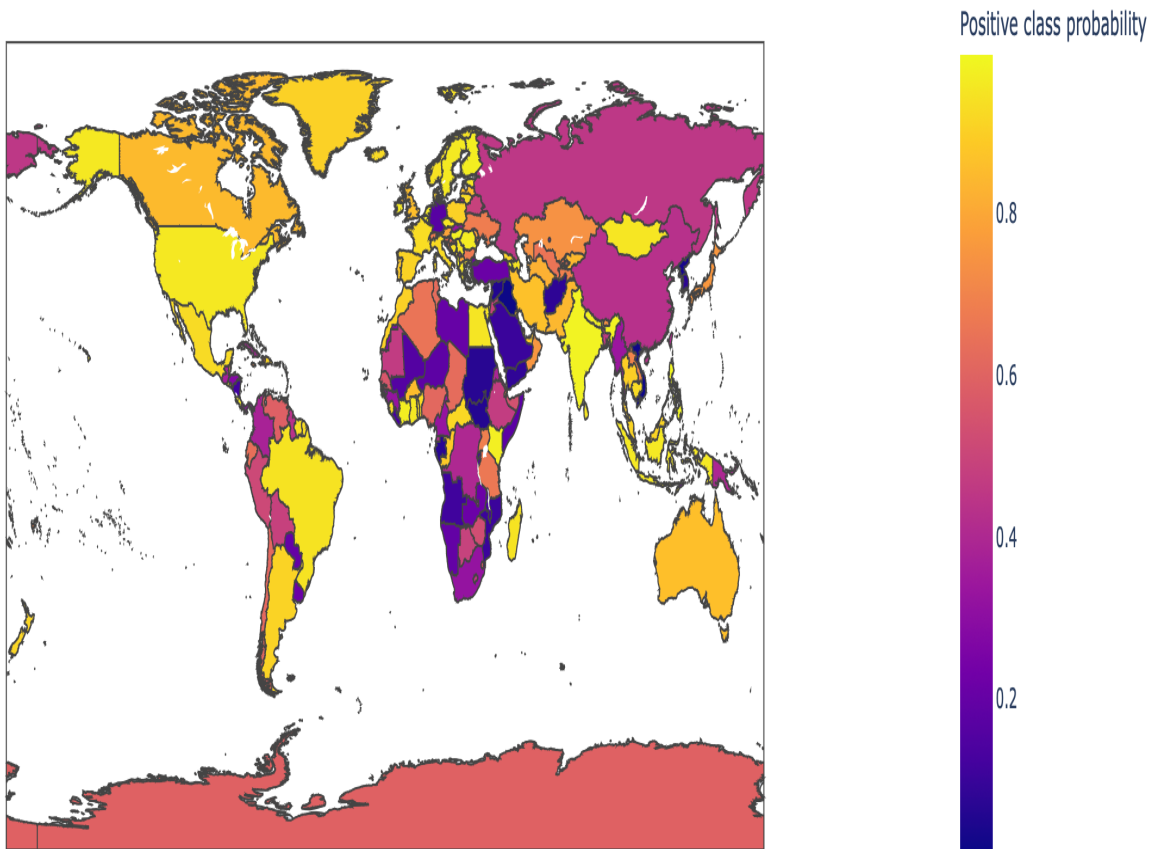
Bias in sentiment analysis

What about biases in sentiment classification models?

- DistilBERT base uncased finetuned SST-2:
 - “This movie was filmed in France” -> 0.89
 - “This movie was filmed in Afghanistan” -> 0.08

Bias in sentiment analysis

- “This movie was filmed in {country_name}”



From Aurélien Geron colab

Summary

Summary

- Sentiment analysis
- Lexicon-based methods
- Supervised methods (traditional ML)
- Deep learning (incl. sentiment-aware word embeddings, BERT)

- Bias

Resources

- Crawl your own data from X/Twitter:
 - <https://docs.x.com/overview>
- SemEval Datasets: 2012-now
 - <https://semeval.github.io/>
- Stanford Sentiment Treebank:
 - <https://nlp.stanford.edu/sentiment/>
- Sanders Corpus:
 - https://github.com/zfz/twitter_corpus
- IMDB movie reviews (50K)
 - <https://ai.stanford.edu/~amaas/data/sentiment/>
- Datasets from Bing Liu’s group:
 - <https://www.cs.uic.edu/~liub/FBS/sentiment-analysis.html>
- Amazon review data
 - <https://nijianmo.github.io/amazon/index.html>
- iSarcasm
 - <https://github.com/dmbavkar/iSarcasm>

Lexicons and tools

- VADER (Hutto and Gilbert, 2014)
 - <https://github.com/cjhutto/vaderSentiment>
- LIWC
 - <https://www.liwc.app/>
- Bing Liu
 - <https://www.cs.uic.edu/~liub/FBS/sentiment-analysis.html>
- SentiWordNet (Esuli and Sebastiani, 2006)
 - <https://github.com/aesuli/SentiWordNet>
- NRC Lexicons
 - <http://saifmohammad.com/WebPages/lexicons.html>
- AFFINN (Nielsen, 2011)
 - <https://github.com/fnielsen/afinn>

Tutorial

- Sentiment analysis in huggingface

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Practical 8